

BALTIMORE HARBOR, MD, USAWMO: **997296**Lat: **39.2669N**Lon: **76.5794W**Elev: **10**StdP: **101.20**Time Zone: **-5.00 (NAE)**Period: **08-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.4	(c) -7.1	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 9.0	(k) -2.5	(l) 8.3	(m) -1.4	(n) 3.7	(o) 330	(p) 0.406

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 7.3	(c) 33.7	(d) N/A	(e) 32.3	(f) N/A	(g) 30.9	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 2.7	(p) 260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 8.1	(b) 7.2	(c) 6.2	(e) -11.7	(f) 36.3	(g) 2.9	(h) 2.0	(i) -13.7	(j) 37.7	(k) -15.4	(l) 38.9	(m) -17.1	(n) 40.0	(o) -19.2	(p) 41.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.5	1.1	2.9	6.9	14.0	19.7	24.1	26.7	25.2	21.9	15.5	9.2	5.2
	DBStd	9.63	4.97	4.87	4.91	4.37	3.96	3.10	2.64	2.11	3.19	3.91	4.12	4.46
	HDD10.0	838	276	204	121	12	0	0	0	0	0	5	61	160
	HDD18.3	2283	533	432	354	143	32	2	0	0	5	103	273	407
	CDD10.0	2465	2	6	26	132	301	424	519	471	357	177	38	12
	CDD18.3	867	0	0	1	13	75	176	261	213	112	17	0	0
	CDH23.3	7102	0	1	4	111	448	1444	2672	1686	664	71	0	0
	CDH26.7	2289	0	0	0	35	110	457	1056	458	161	12	0	0
Wind	WSAvg	2.7	2.8	2.9	3.1	2.9	2.6	2.6	2.3	2.4	2.6	2.7	2.7	2.7
Precipitation	PrecAvg	1067	78	79	98	79	100	78	96	105	92	75	88	94
	PrecMax	1564	220	176	248	192	236	254	297	377	223	220	222	205
	PrecMin	751	10	17	24	9	15	16	18	8	2	3	11	18
	PrecStd	223	44	42	54	42	52	45	54	76	61	48	49	56
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.7	20.3	23.3	30.9	31.8	34.6	36.5	33.3	32.4	28.0	21.3	19.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	12.8	15.6	19.7	26.5	29.6	32.7	34.7	31.6	30.3	25.3	18.7	16.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	10.1	11.9	16.9	23.2	27.6	30.9	33.0	30.4	28.6	23.0	16.9	14.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	8.0	9.3	14.3	20.6	25.8	29.3	31.4	29.2	26.9	21.3	15.3	11.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	6.8	7.1	7.5	8.6	7.7	7.2	7.3	6.7	6.7	7.0	7.0	6.2
		MCDBR	10.3	11.4	11.8	12.5	9.7	9.3	9.0	8.1	8.2	9.2	9.2	9.3
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub		0.311	0.322	0.362	0.401	0.446	0.500	0.518	0.513	0.429	0.364	0.326	0.311
	taud		2.467	2.422	2.374	2.287	2.194	2.085	2.050	2.048	2.255	2.443	2.507	2.507
	Ebn at Noon		861	901	893	875	839	792	772	764	812	840	838	835
	Edh at Noon		80	95	111	129	144	161	166	162	124	93	76	72
All-Sky Solar Radiation	RadAvg		2.03	2.88	3.82	4.91	5.49	6.09	6.03	5.34	4.35	3.22	2.32	1.78
	RadStd		0.17	0.33	0.28	0.40	0.66	0.47	0.29	0.30	0.51	0.35	0.20	0.17

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (79 neighbors)		+0.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+129	+73	

Nomenclature: See separate page